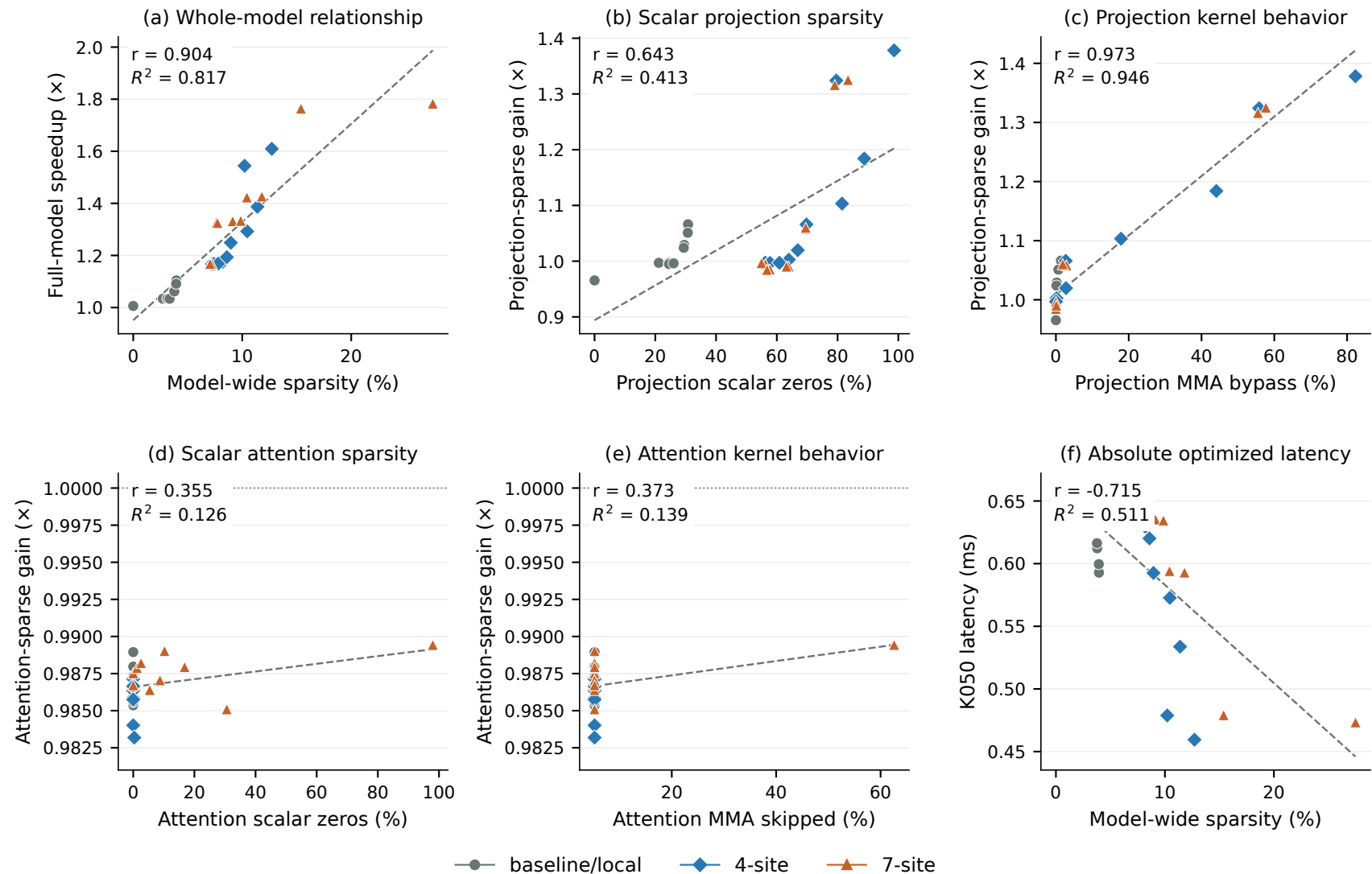
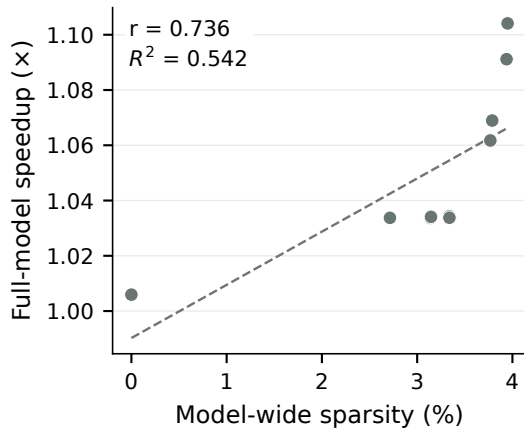


K050: scalar sparsity, kernel counters and runtime gains

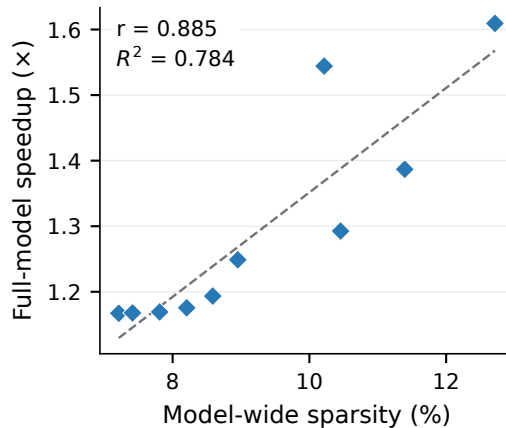


The sparsity-speedup relationship also exists within recipe families

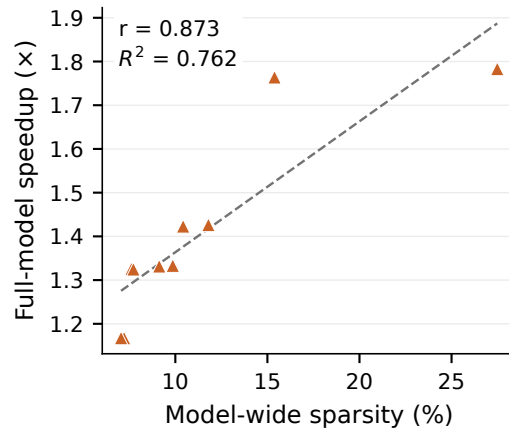
baseline/local (n = 10)



4-site (n = 10)



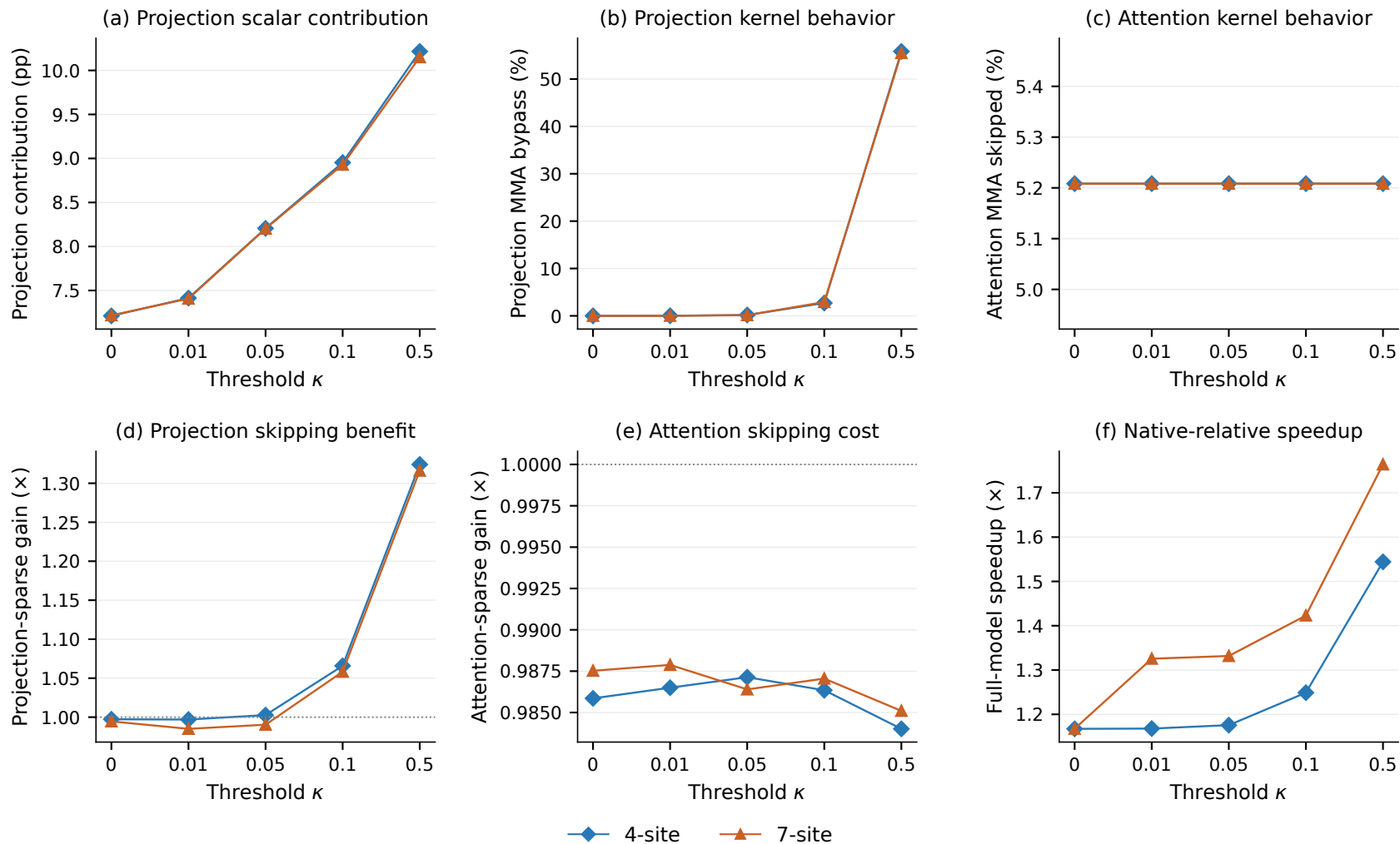
7-site (n = 10)



Family means alone: $R^2 = 0.499$; after within-family centering: $R^2 = 0.660$.

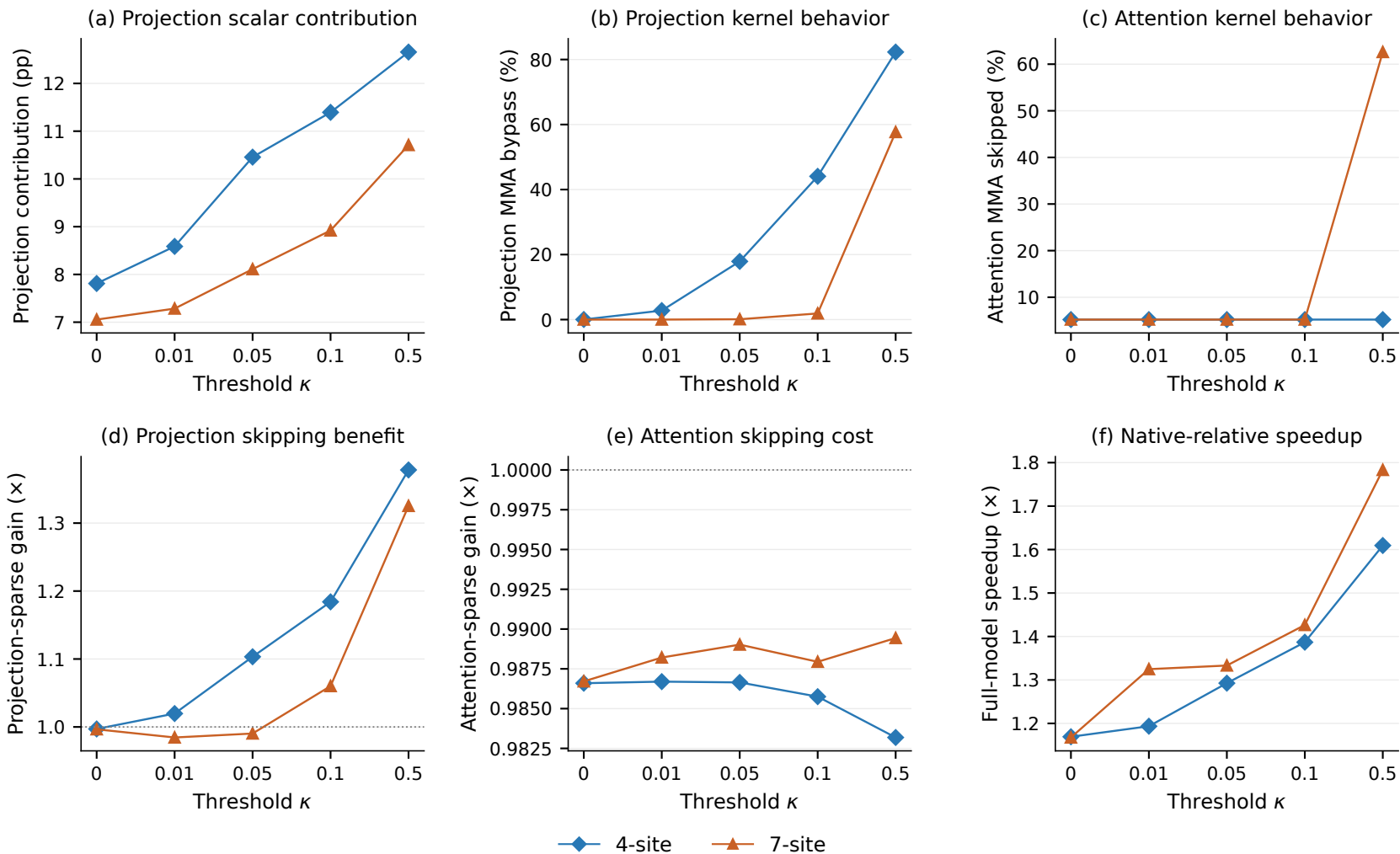
Axis ranges vary to expose within-family variation. Descriptive fits across different trained checkpoints; no causal interpretation.

Matched 4-site versus 7-site recipes without pressure



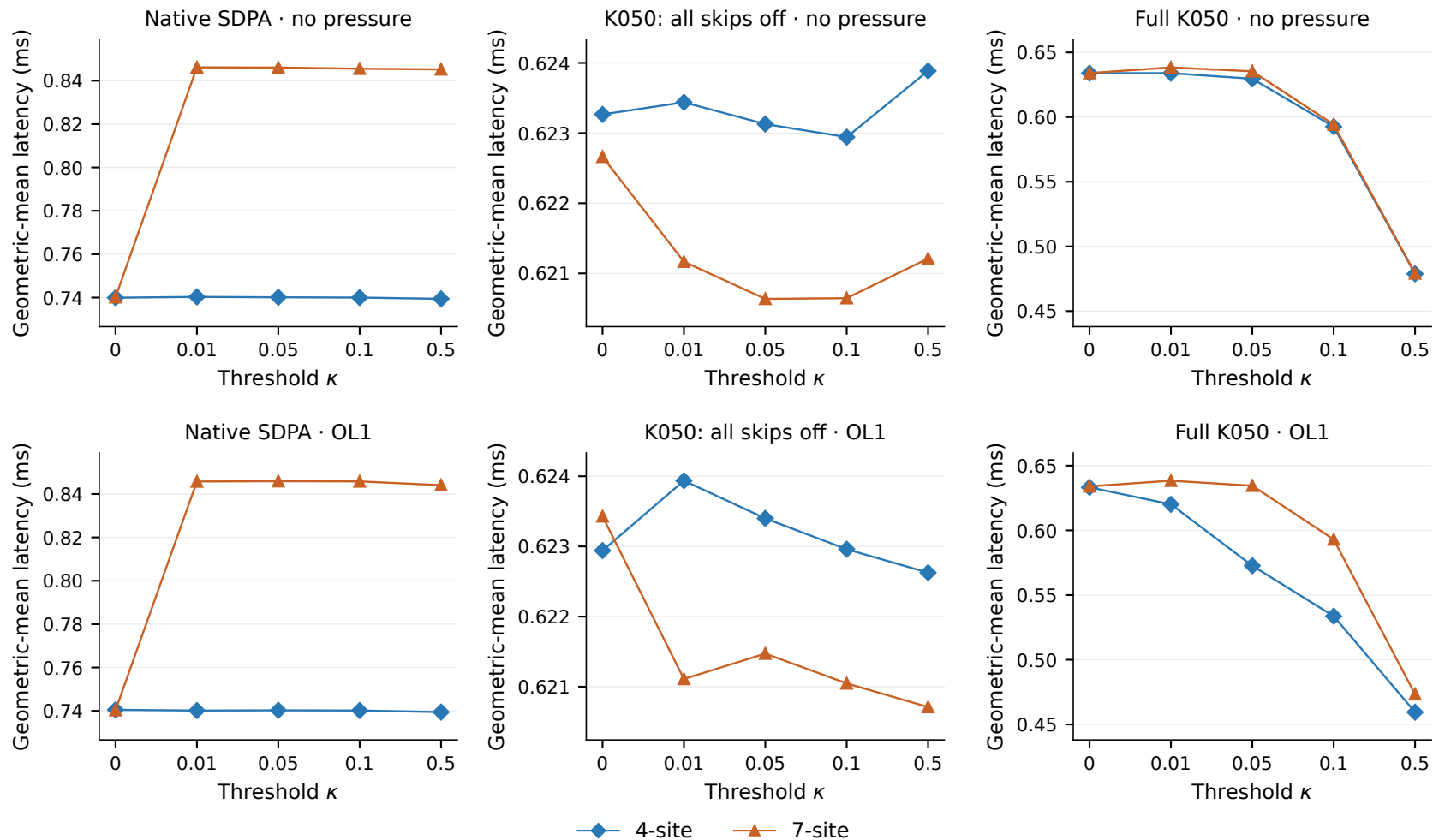
Same threshold and pressure method. Projection gain = no-skip / attention-dense latency; attention gain = attention-dense / full latency.

Matched 4-site versus 7-site recipes with OLI



Same threshold and pressure method. Projection gain = no-skip / attention-dense latency; attention gain = attention-dense / full latency.

Higher native-relative speedup does not mean lower K050 latency



Same 64 inputs $\times$ 7 passes $\times$ 3 processes per implementation; columns use different latency ranges, shared between pressure settings.